

Unit 2 – Exploring Two Variable Data

Calculator Commands Summary Sheet

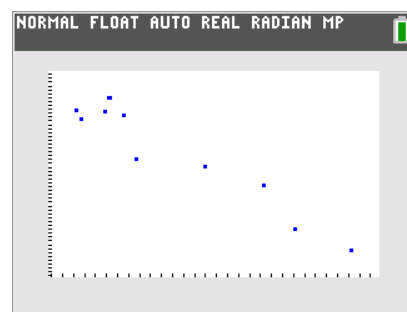
Creating a Scatterplot on the Calculator

- 1) Load the x-values into List 1 and the y-values into List 2.
- 2) Using StatPlot - highlight the mini-scatterplot: XList: L1 and YList: L2
- 3) Press “graph” but you will have to “Zoom 9” to fit the scatterplot on the screen.

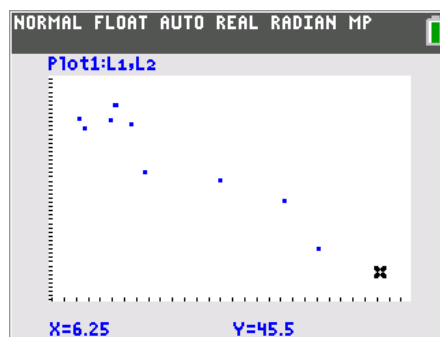
L1	L2	L3	L4	L5	1
6.25	45.5	-----	-----	-----	
5.22	51.2				
2.11	80.5				
1.86	85				
1.85	84.9				
4.66	62.3				
2.35	69				
1.78	81.5				
3.58	67.3				
1.33	79.5				
1.25	81.6				

L1={6.25, 5.22, 2.11, 1.86, 1.8

Plot1	Plot2	Plot3
On	Off	
Type:		
Xlist: L1		
Ylist: L2		
Mark:		
Color: BLUE		



Pressing “Trace” will take you through all the ordered pairs, in the order they were put in the list.



Turning STAT Diagnostic On

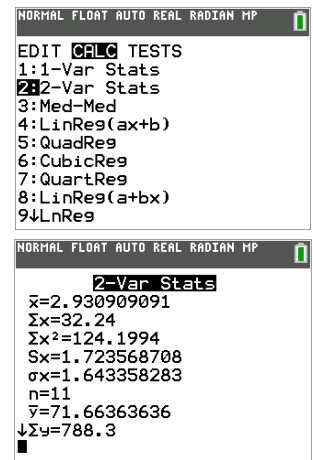
You will need to turn STAT Diagnostic On if you are using a calculator for the first time. This is something that only needs to be done once, and you need it to have your correlation show up.

MODE	Diagnostic On	Diagnostic Off
NORMAL FLOAT AUTO REAL RADIAN MP DISPLAY CORR COEFF (r, r ² , R ²) MATHPRINT CLASSIC NORMAL SCI ENG FLOAT 0 1 2 3 4 5 6 7 8 9 RADIAN DEGREE FUNCTION PARAMETRIC POLAR SEQ THICK DOT-THICK THIN DOT-THIN SEQUENTIAL SIMUL REAL a+bt re^(ti) FULL HORIZONTAL GRAPH-TABLE FRACTION TYPE: Un/d ANSWERS: AUTO DEC STAT DIAGNOSTICS: OFF ON STAT MATH OPS: ON OFF SET CLOCK 01/01/15 12:00 AM LANGUAGE: ENGLISH	NORMAL FLOAT AUTO REAL RADIAN MP LinReg y=ax+b a=-7.660499951 b=94.11586531 r ² =0.9150467667 r=-0.9565807685	NORMAL FLOAT AUTO REAL RADIAN MP LinReg y=ax+b a=-7.660499951 b=94.11586531

2 Variable Statistics

This is a great feature to run when you have x and y values and need summary statistics for both of them. Alternatively, you could run 1-Var Stats on each list, but why do that twice when you can do this once!

- 1) Load the x-values into List 1 and the y-values into List 2
- 2) Stat → Calc → 2: 2-VarStats
- 3) Put XList as L1 and YList as L2 and leave FreqList blank
- 4) Scroll up and down to see all the summary statistics for each list



Least Squares Regression Line

There are two LinReg options on your calculator, and the difference is just the format you want your regression line to be in; they are both calculated the same way.

Stat → Calc 4:LinReg(ax + b)	Form: $\hat{y} = ax + b$ $\hat{y} = -7.66x + 94.12$		
Stat → Calc 8:LinReg(a + bx)	Form: $\hat{y} = a + bx$ $\hat{y} = 94.12 - 7.66x$		

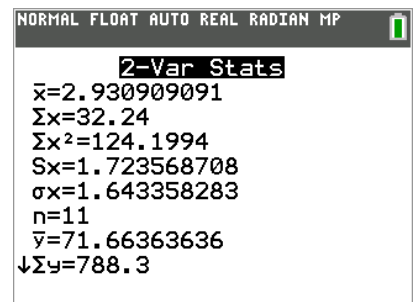
You can “Store Reg Equ” if you want to graph the LSRL on your scatterplot on your calculator. You can leave this blank if you do not care about that.

Find Correlation by Hand

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

You will NOT need to know this for our exams (or the AP Exam) but here is a reference for how we would do this.

- 1) Enter x values into L1 and y values into L2
- 2) Find the following summary statistics using 2-Var Stat: s_y , s_x , $\bar{x}$, $\bar{y}$



3) Do some list magic!

Highlighting your L3 list title, you will find the z-scores of the x-values:

NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	3
6.25	45.5	-----	-----	-----	
5.22	51.2				
2.11	80.5				
1.86	85				
1.85	84.9				
4.66	62.3				
2.35	69				
1.78	81.5				
3.58	67.3				
1.33	79.5				
1.25	81.6				

$L3=(L1-2.93)/1.72$

Highlighting your L4 list title, you will find the z-scores of the y-values:

NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	4
6.25	45.5	1.9302	-----	-----	
5.22	51.2	1.3314			
2.11	80.5	-0.477			
1.86	85	-0.622			
1.85	84.9	-0.628			
4.66	62.3	1.0058			
2.35	69	-0.337			
1.78	81.5	-0.669			
3.58	67.3	0.3779			
1.33	79.5	-0.93			
1.25	81.6	-0.977			

$L4=(L2-71.66)/13.8$

Highlighting your L5 list title, you will multiply L3 and L4 together:

NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	5
6.25	45.5	1.9302	-1.896	-----	
5.22	51.2	1.3314	-1.483		
2.11	80.5	-0.477	0.6406		
1.86	85	-0.622	0.9667		
1.85	84.9	-0.628	0.9594		
4.66	62.3	1.0058	-0.678		
2.35	69	-0.337	-0.193		
1.78	81.5	-0.669	0.713		
3.58	67.3	0.3779	-0.316		
1.33	79.5	-0.93	0.5681		
1.25	81.6	-0.977	0.7203		

$L5=L3 * L4$

4) Perform 1-Var Stat on L5 to find the sum of the list ($\sum x$) this gives you the value of $\sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$

NORMAL FLOAT AUTO REAL RADIAN MP	
1-Var Stats	1-Var Stats
List:L5	$\bar{x} = -0.8715935901$
FreqList:	$\Sigma x = -9.587529491$
Calculate	$\Sigma x^2 = 19.58831449$
	$Sx = 1.05980589$
	$\sigma x = 1.010485268$
	$n = 11$
	$\min X = -3.659049545$
	$\downarrow Q1 = -0.7035389282$

5) Calculate r using n as the total number of observation and your sum from part 4.

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) = \frac{1}{11-1} (-9.5875) = -0.95875$$

Residual plots

- The calculator will generate a complete set of residuals when you perform LinReg.
- RESID stores only the current set of residuals; a new set of residuals is stored in RESID each time you perform a new regression.

- Enter your x values in L1 and y values in L2
- Perform a LinReg, either using 4:LinReg(ax + b) or 8:LinReg(a+bx)
- Go to Stat Plot, XList is L1 and YList is RESID (get this by pressing 2nd → STAT → 9:RESID)
- Zoom 9:Stat and you will have your plot!
- Pressing "Trace" will give you the x-value and the corresponding residual value for "y"

NORMAL FLOAT AUTO REAL RADIAN MP	
NAMES	OPS MATH
1:L1	
2:L2	
3:L3	
4:L4	
5:L5	
6:L6	
7:RESID	

NORMAL FLOAT AUTO REAL RADIAN MP	
Plot1	Plot2 Plot3
On	Off
Type:	
Xlist:L1	
Ylist:RESID	
Mark: $\square + \blacksquare$	
Color: BLUE	

